



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ARUN ARVIND AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 82.99 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.651594, 78.466362)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map The route from CVQF+23W in Sangagiri to MF28+HRW in Puthiragoundampalayam spans approximately 83 kilometers. The path mainly follows NH544, a major national highway, before transitioning to state and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards Tamil Nadu experiences a tropical climate, with high temperatures and humidity throughout the year. The monsoon season (June to September) can lead to heavy rainfall, resulting in waterlogging and decreased visibility. Dust storms in the dry months and occasional cyclonic winds could also pose hazards.

3. Traffic Patterns and Congestion-Prone Areas The route intersects several towns including Rasipuram and Erode, which experience moderate to heavy traffic, especially during peak hours (8-10 AM and 5-7 PM). Expect increased congestion in these town centers and near any major junctions or toll booths.

4. Road Quality and Infrastructure Assessment National highways in the region are generally well-maintained, offering smooth travel. However, some sections of rural roads towards Puthiragoundampalayam are narrower and may be prone to potholes, especially post-monsoon.

5. Suggestions for Alternative Routes for Emergencies In case of a roadblock on NH544, consider bypassing through State Highway 86 to avoid congestion. Always have a backup plan to reroute through state highways that run parallel to your main route, such as SH89.

6. Local Regulations Affecting Hazardous Material Transport Tamil Nadu has specific regulations regarding the transport of hazardous materials: secure permits, adhere to designated routes, and maintain appropriate signage on the vehicle. Compliance is crucial to avoid fines and ensure safety.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials Historically, accidents involving heavy vehicles in the region have been attributed to overspeeding and poor road conditions. Major incidents involved overturning due to sharp turns or slippery roads during the rainy season. Awareness of previous incidents can guide precautionary measures.

8. Environmental Considerations and Sensitive Areas The route skirts agricultural zones and small water bodies. Avoid contamination by adhering to spill-prevention measures. Noise and air pollution are additional concerns in densely populated sections.

9. Communication Coverage and Potential Dead Zones The majority of the route retains strong mobile network connectivity thanks to infrastructure along NH544. Rural stretches nearing Puthiragoundampalayam may experience intermittent coverage issues, particularly in low-lying or remote areas.

10. Estimated Emergency Response Times for Different Route Segments Emergency services along NH544 have an average response time of 15-30 minutes due to proximity to major towns. Rural areas may face longer response times, up to 45 minutes, due to limited accessibility and resources.

11. Overall Summary of Risk Assessment The primary risks involve weather-related road conditions, traffic congestion, and varying road infrastructure quality. Awareness and preparation, especially during monsoon seasons, are vital. Ensuring proper compliance with regulations and bearing in mind potential connectivity issues can significantly mitigate risks associated with transporting hazardous materials.

This comprehensive route analysis emphasizes the importance of safety measures and preparation to minimize hazards encountered during the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.64140, 78.11801	30 KM/Hr
13	Turn	Medium	11.64179, 78.11822	30 KM/Hr
14	Turn	Medium	11.64218, 78.11881	30 KM/Hr
15	Turn	Medium	11.64192, 78.12133	30 KM/Hr
16	Turn	Medium	11.64156, 78.12162	30 KM/Hr
17	Turn	Medium	11.64116, 78.12168	30 KM/Hr
18	Turn	Medium	11.63934, 78.12082	30 KM/Hr
19	Turn	Medium	11.65154, 78.35007	30 KM/Hr
20	Blind Spot	Blind Spot	11.65111, 78.47828	10 KM/Hr
21	Turn	High	11.65074, 78.47818	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
24	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
26	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
27	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
28	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
29	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
32	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
33	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
40	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
42	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
34	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
35	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
36	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium